

Course Title	Data Science
Course Code	EI-334
Credit Hours	3
Category	Technical Elective
Prerequisite	DI-326: Artificial Intelligence
Co-Requisite	None
Follow Up	None
Course Description	Python Programming: Environment, python programming techniques such as lambdas, reading and manipulating csv files, and the numpy library. Data Manipulation and Cleaning: Python Pandas data science library, abstraction of the Series and DataFrame as the central data structures for data analysis, functions such as groupby, merge, and pivot tables. Inferential Statistical Analyses. Machine learning vs Descriptive Statistics, scikit learn toolkit, data dimensionality, clustering, evaluating clusters. Supervised Learning: predictive modelling, data generalizability, cross validation, Overfitting.
Text Book(s)	1. Joel Grus, Data Science from Scratch: First Principles with Python, 1st Edition, O'Reilly, 2015, ISBN: 1-491-90142-X.
Reference Material	1. https://www.coursera.org/specializations/data-science-python 2. https://www.coursera.org/learn/python-data-analysis/home/info 3. https://www.coursera.org/degrees/master-of-computer-science-asu 4. https://www.coursera.org/degrees/master-of-applied-data-science-umich 5. https://www.coursera.org/degrees/masters-in-computer-data-science

3. INFORMATION TECHNOLOGY DOMAIN CORE COURSES

Course Title	IT Project Management
Course Code	DI-321
Credit Hours	3
Category	Domain Core
Prerequisite	None
Co-Requisite	None
Follow Up	None
Course Description	Introduction: Project Management. Project Management in Information Technology Context. The Project Management Process Groups. Scope triangle, The S curve. Five Phases of Project Management Life Cycle: Defining, planning, executing, controlling, closing, Scope of project, Project Work Breakdown Structure (WBS), Project Time Management. Estimate Activity Duration: Causes of variation in activity duration, five methods of Estimating Activity Duration. Project Networks: Critical Path Method (CPM), Build the project network, Analysis of the project network, Project Evaluation Review Technique (PERT), Activity on Arrow, Activity on Node, GANTT Chart, Using MS-Project to draw GANTT chart and project Networks. Project Proposal: Purpose of the project proposal, Contents of the project proposal, Format of the Project Proposal. Staffing and Personnel Planning: Personnel Plan, Team structures, Democratic decentralized, Controlled Decentralized, Controlled Centralized, Coordination and communication issues, Maslow's need theory of motivation, Software Project Team Roles and Responsibilities, Risk Identification, Analysis and Management. Project Management: Integration Management. Scope Management. Cost Management. Quality Management. Human Resource Management. Communications Management. Risk Management. Procurement Management. Management Tools. Configuration Management: Change Management and Control, baselines, Version Control, Earned Value Analysis for Project Monitoring and Control. Examples of Earned Value Analysis, Project Quality Assurance Plans, SQA Process Project Quality Standards, Project Documents.
Text Book(s)	1. Guide to the Project Management Body of Knowledge (PMBOK® Guide), 6 th Edition, Project Management Institute, 2017, ISBN-10: 1628251840.
Reference Material	1. Teresa Luckey, Joseph Phillips, Software Project Management for Dummies, 1st Edition, For Dummies Publisher, 2006, ISBN 0471749346.

Course Title	Cyber Security
Course Code	DI-325
Credit Hours	3
Category	Domain Core
Prerequisite	
Co-Requisite	None
Follow-up	None
Course Description	Topics: Basic security concepts, Information security terminology, Malware classifications, Types of malware. Server-side web applications attacks. Cross-site scripting, SQL Injection, Cross-site request forgery, Planning and policy, Network protocols and service models. Transport layer security, Network layer security, Wireless security, Cloud & IoT security.
Text Book(s)	1. Mark Ciampa, Security+ Guide to Network Security Fundamentals, 4th Edition.
Reference Material	1. Randall J. Boyle, Corporate Computer Society, 3rd Edition.

Course Title	Database Administration and Management Lab
Course Code	DI-324L
Credit Hours	1
Category	Domain Core
Prerequisite	CC-312: Database Systems
Co-Requisite	None
Follow-up	None
Course Description	Introduction: to Enterprise Database Systems. Oracle Architectural Components: What is Introduction to Enterprise Database Systems, Responsibilities of Database Administrator. Oracle Server; Oracle Instance; Physical Structure. Creating Database and Data dictionary: Creation Prerequisites; Creating Database; Starting Instance. Data Dictionary; Data Dictionary Contents. Dynamic Performance Views. Managing Control Files and Redo Log Files: Control File; Control File Contents; Multiplexing Control File; Using Redo Log Files; Structure of Redo Log File; Adding Redo Log Groups and Members; s. Managing Tablespaces: Database Storage Hierarchy; Kinds of Tablespaces; Managing Data files. Storage Structure and Relationships: Adding Data files and other Operations with Data files; Types of Segments. Managing Undo Data: Undo Segment; Read Consistency; Automatic undo Management: Managing Tables. Managing Users: Creating Users; Managing Users; Creating Tables; Managing Tables. Managing Indexes: types of Indexes; Creation and Management of Indexes. Maintaining Data Integrity: Types of Constraints; Managing Constraints. Managing Privileges: System Privileges; Object Privileges. Managing Roles: Assign and Revoke Roles and Roles Management. Network overview, Configuring Listener; Sessions. Client-Side Configuration: Host Naming Method; Local Naming Method; Net Assistant; Configurations of Different Methods. Usage and Configuration of Oracle Shared Server: Server Configuration; Dedicated Server Process; Shared Server Process. Backup and Recovery: overview, Instance and Media Recovery, Categories of Failure; Defining Backup and Recovery Strategy. Configuration of Archive log mode: Switching to Archive Log Mode; Archive Process. User Managed Backups: User Managed backups and Issues. User Managed Complete Recovery: How To recover a Database Manually Using Manual Backups. User Managed Incomplete Recovery: How to DO an Incomplete Recovery. Import/Export: Different export and import modes.
Text Book(s)	1. Oracle 11i DBA Fundamentals I (Student Guide, Volume I and II). 2. Oracle 11i DBA Fundamentals II (Student Guide, Volume I and II). 3. Performance Tuning (Student Guide, Volume I and II).
Reference Material	1. Oracle Magazine. 2. Online Material URL http://otn.oracle.com/

Course Title	Database Administration and Management
Course Code	DI-324
Credit Hours	3+1
Category	Domain Core
Prerequisite	CC-312: Database Systems
Co-Requisite	None
Follow-up	None
Course Description	Introduction to advanced data models such as object relational, object oriented. File organization concepts, Transactional processing and Concurrency control techniques, Recovery techniques, Query processing and optimization, Database Programming, Integrity and security, Database Administration, Physical database design and tuning, Distributed database systems, Emerging research trends in database systems.
Text Book(s)	1. Ramez Elmasri, Shamkant Navathe, Fundamentals of Database Systems, 6 th Edition, Addison Wesley, 2010, ISBN: 0136086209.
Reference Material	1. Henry F. Korth, Abraham Silberschatz, Database System Concepts, 6 th Edition, McGraw Hill, 2010, ISBN: 0073523321.

UNIVERSITY OF THE PUNJAB

NOTIFICATION

It is hereby notified that the Vice Chancellor has approved the following recommendations/decisions made in a meeting of Centre for Quran and Sunnah chaired by the Pro-Vice Chancellor:-

کریڈٹ آورز (Credit Hours) کی تقسیم:

یونیورسٹی نوٹیفیکیشن کی روشنی میں بی ایس (آنرز) کے لیے سالانہ ایک کریڈٹ آور (Credit Hour) کرلیب ورک کریڈٹ آور کے مساوی قرار دیا گیا ہے۔ اور اس طرح ہر سمسٹر میں ہفتہ وار تین کلاسیں پڑھائی جائیں گی۔

پہلا، تیسرا، پانچواں اور ساتواں سمسٹر نان کریڈٹ آور (Non Credit Hour) تصور کیا جائے گا۔ جبکہ دوسرا، چوتھا، چھٹا اور آٹھواں سمسٹر (ہر سمسٹر) ایک کریڈٹ آور کے مساوی ہوگا۔

اس سیکٹ کی assessment باقی مضامین کی طرح ہوگی اور ان کے مائیکس، میڈیم اور فائنل مسٹرز باقی امتحان بھی باقی مضامین کی طرح ہونگے۔
نصاب:

الف۔ بی ایس آنرز :

سال اول : سمسٹر I - سورة الفاتحة سورة ال عمران -

سمسٹر II - سورة النساء سورة الانعام -

سال دوم : سمسٹر III - سورة الاعراف سورة يونس -

سمسٹر IV - سورة هود سورة الكهف -

سال سوم : سمسٹر V - سورة مريم سورة النور -

سمسٹر VI - سورة الشعراء سورة ص -

سال چہارم : سمسٹر VII - سورة الزمر سورة ق -

سمسٹر VIII - سورة الذاريات سورة الناس -

ب۔ ایم اے/ایم ایس سی : بی ایس آنرز کے سال سوم اور چہارم کا نصاب

ج۔ ایم فل : سورة المؤمنون ، سورة الاحزاب اور سورة الحشر کا ترجمہ

دہی ایچ ڈی : سورة النور ، سورة الفرقان اور سورة الحجرات

نوٹ : i. ایم اے، ایم فل اور پی ایچ ڈی کا کورس ایک کریڈٹ آور (Credit Hour) پر مشتمل ہوگا۔

ii. بی ایس آنرز کے پہلے دو سال کا نصاب ADP میں پڑھایا جائے گا۔

تدریس ترجمہ قرآن کورس، بی ایس آنرز کے پہلے سمسٹر سے اٹھویں سمسٹر کے لیے
مندرجہ ذیل کورس کوثر ترتیب دیے ہیں۔

HQ-001	تدریس ترجمہ قرآن	سمسٹر I-	سال اول:
HQ-002	تدریس ترجمہ قرآن	سمسٹر II-	
HQ-003	تدریس ترجمہ قرآن	سمسٹر III-	سال دوم:
HQ-004	تدریس ترجمہ قرآن	سمسٹر IV-	
HQ-005	تدریس ترجمہ قرآن	سمسٹر V-	سال سوم:
HQ-006	تدریس ترجمہ قرآن	سمسٹر VI-	
HQ-007	تدریس ترجمہ قرآن	سمسٹر VII-	سال چہارم:
HQ-008	تدریس ترجمہ قرآن	سمسٹر VIII-	

Admin. Block,
Quaid-i-Azam Campus,
Lahore.

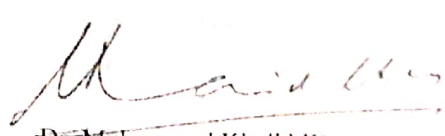
Sd/-
Dr. Muhammad Khalid Khan
Registrar

No. D/330/Acad.

Dated: 05-01-2021

Copy of the above is forwarded to the following for information and further
necessary action:-

1. Pro-Vice Chancellor
2. Deans of Faculties
3. Heads of University Teaching Departments
4. Secretary to the Vice-Chancellor
5. Private Secretary to Registrar
6. Notification file


Dr. Muhammad Khalid Khan
Registrar

UNIVERSITY OF THE PUNJAB
NOTIFICATION

It is hereby notified that the Syndicate in its meeting held on 18.08.2020 approved the recommendations of the Academic Council dated 24.06.2020 regarding teaching of Holy Quran with translation under a separate centre "Centre for Quran and Sunnah", as per following guidelines proposed by the Seven-Members Committee of Vice Chancellors constituted by the honourable Chancellor/Governor Punjab:-

1. The teaching of translation of the Holy Quran may be relevant to the profession, faith, ethics & morality and practices (ایمان، عبادت، معاملات، اخلاقیات و عادات).
2. The teaching of the Holy Quran may be undertaken with "translation only".
3. The Quran course may have equivalent to one credit hour per year (Theory/Practical), as the relevant University feels appropriate. However, the HCE limit of total credit hours in Undergraduate Program must be followed.
4. The Quran course may be taught in addition to the HEC-approved compulsory Pakistan and Islamic studies.
5. The assessment of the Quran course must require each candidate to pass/fail. The student may be further assigned a grade as in any other credited subject, if University feels necessary/appropriate.
6. The Quran course is mandatory/compulsory and may be spread in a manner to engage students throughout the duration of the course.
7. As a substitute, the non-Muslims may be offered the course of Ethics as is already the case in many universities.
8. Universities may make a concrete effort to generate interest in this compulsory course so that the course is taught in the best spirit of drawing learnings from the Holy Quran.
9. A suggested list of Quran Translations (Urdu & English) is attached.

Quaid-i-Azam Campus,
Lahore.

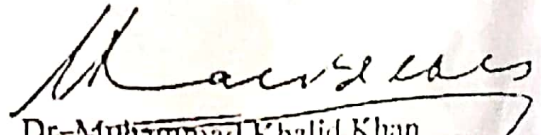
Sd/-
Dr. Muhammad Khalid Khan
Registrar

No. D/5477/Acad.

Dated: 01-09-2020.

Copy of the above is forwarded to the following for information and further necessary action:-

1. Pro-Vice Chancellor
2. All the Deans of the Faculties
3. All the Heads of the University Teaching Departments
4. Controller of Examinations
5. Director, IT Centre
6. Deputy Registrar (Affiliation)
7. Secretary to the Vice-Chancellor
8. Secretary to the Registrar
9. Admin. Officer (Statutes)
10. Admin. Officer (Senate/Syndicate)
11. Admin. Officer (Meetings-II)


Dr. Muhammad Khalid Khan

Model Paper for Affiliated Colleges



Course:
Program:
Duration:
Paper Date:

Department:
Exam:

Translation of Holy Quran
BS 4 Year Degree Program
Two Hours

Final Term Exam.

Course Code: HQ-001
Semester: Spring/Fall 20--
Total Marks: 60
Page(s): 3
Student's Name: _____
Roll No: _____

Instruction/Notes:

Attempt this Paper on the Same Sheet.

20 نمبر

سؤال 1 : مندرجہ ذیل آیات مبارکہ کا اردو/انگریزی میں ترجمہ کریں۔

Translate the Following Verses into Urdu or English Language

1- حَافِظُوا عَلَى الصَّلَوَاتِ وَالصَّلَاةِ الْوُسْطَى وَقُومُوا لِلَّهِ قَانِتِينَ.

2- وَقَاتِلُوا فِي سَبِيلِ اللَّهِ وَعَلِمُوا أَنَّ اللَّهَ سَمِيعٌ عَلِيمٌ.

3- قَالَ إِنَّ اللَّهَ اصْطَفَاهُ عَلَيْكُمْ وَزَادَهُ بَسْطَةً فِي الْعِلْمِ وَالْجِسْمِ وَاللَّهُ يُؤْتِي مَلَكُهُ مَن يَشَاءُ ۚ وَاللَّهُ وَاسِعٌ عَلِيمٌ

4- قَوْلٌ مَّعْرُوفٌ وَمَغْفِرَةٌ خَيْرٌ مِّنْ صَدَقَةٍ يَتْبَعُهَا أَذًى ۗ وَاللَّهُ غَنِيٌّ حَلِيمٌ

5- يَا أَيُّهَا الَّذِينَ آمَنُوا لَا تُبْطِلُوا صَدَقَاتِكُمْ بِالْمَنِّ وَالْأَذَى كَالَّذِي يُنْفِقُ مَالَهُ رِثَاءَ النَّاسِ وَلَا يُؤْمِنُ بِاللَّهِ وَالْيَوْمِ الْآخِرِ ۚ

6- نَزَّلَ عَلَيْكَ الْكِتَابَ بِالْحَقِّ مُصَدِّقًا لِّمَا بَيْنَ يَدَيْهِ وَأَنزَلَ التَّوْرَةَ وَالْإِنْجِيلَ

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University of the Punjab, Quaid-e-Azam Campus, Lahore

7- رَبَّنَا لَا تُزِغْ قُلُوبَنَا بَعْدَ إِذْ هَدَيْتَنَا وَهَبْ لَنَا مِنْ لَدُنْكَ رَحْمَةً إِنَّكَ أَنْتَ الْوَهَّابُ

8- تَوَلَّجَ اللَّيْلُ فِي النَّهَارِ وَتَوَلَّجَ النَّهَارُ فِي اللَّيْلِ وَتُخْرِجُ الْحَيَّ مِنَ الْمَيِّتِ وَتُخْرِجُ الْمَيِّتَ مِنَ الْحَيِّ وَتَرْزُقُ مَنْ تَشَاءُ بِغَيْرِ حِسَابٍ

9- وَادْكُرْ رَبَّكَ كَثِيرًا وَسَبِّحْ بِالْعَشِيِّ وَالْإِبْكَارِ

10- وَمَنْ يَبْتَغِ غَيْرَ الْإِسْلَامِ دِينًا فَلَنْ يُقْبَلَ مِنْهُ وَهُوَ فِي الْآخِرَةِ مِنَ الْخَاسِرِينَ

10 نمبر

سؤال 2: ”رب“ کے ساتھ ضمائر متصلہ لگائیں اور ان کے معانی بتائیں۔

Use attached pronouns with ”رب“ and give their meanings.

- | | | | |
|-------|-------|-------------------|----|
| | _____ | = رَبُّ + هُ = | -1 |
| | _____ | = رَبُّ + هُمْ = | -2 |
| | _____ | = رَبُّ + هَا = | -3 |
| | _____ | = رَبُّ + هُنَّ = | -4 |
| | _____ | = رَبُّ + كُ = | -5 |
| | _____ | = رَبُّ + كُمْ = | -6 |

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- = رَبُّ + كِ = -7
- = رَبُّ + كُنْ = -8
- = رَبُّ + ي = -9
- = رَبُّ + نَا = -10

10 نمبر

سوال 3 : مندرجہ ذیل قرآنی الفاظ کے اردو/انگریزی میں معانی لکھیں۔

Write the meanings of the following Qur'anic words

- | | | | | | |
|-------|-------------|-------|---------|-------|-----------|
| -1 | فِتْنَةٍ | -2 | سَبِيلٍ | -3 | ذَهَبٍ |
| ----- | ----- | ----- | ----- | ----- | ----- |
| -4 | فُضَّةٍ | -5 | نِسَائٍ | -6 | اِنْخَاڑَ |
| ----- | ----- | ----- | ----- | ----- | ----- |
| -7 | الْمَلَكَةِ | -8 | قَسَطٍ | -9 | سَمِيعٍ |
| ----- | ----- | ----- | ----- | ----- | ----- |
| -10 | ذُرِّيَّةٍ | -11 | عَاوِرٍ | -12 | مَهْدٍ |
| ----- | ----- | ----- | ----- | ----- | ----- |
| -13 | يَنْبُرُكٍ | -14 | طَيْرٍ | -15 | يَنْتٍ |
| ----- | ----- | ----- | ----- | ----- | ----- |
| -16 | تَاكُلُونَ | -17 | نَحْلٍ | -18 | الَّذِينَ |
| ----- | ----- | ----- | ----- | ----- | ----- |
| -19 | طِفَامٍ | -20 | بُرٍ | | |
| ----- | ----- | ----- | ----- | | |

10 نمبر

سوال 4 : مندرجہ ذیل الفاظ کی مدد سے واحد کی جمع لکھیں۔

Write the plural with the given words.

أُولَئِكَ - الَّذِينَ - مَوَاتٍ - هُمْ - يَنْتَ - إِلَهٌ - نَبِيٌّ - رَسُلٌ - كُتُبٌ - غَافِلُونَ - آيَاتٌ -

يُؤْتِ - ثَوَابٌ - مُحْسِنِينَ - أَلْبَابٌ - أَعْمَالٌ - يَتْلُو - مُؤْمِنُونَ - تَوَابِينَ - أَصْحَابٌ -

ذَلِكَ	هُوَ	يَنْتَ	كِتَابٌ	بَابٌ	
الَّذِي	يَنْتَ	إِلَهٌ	غَافِلٌ	مُحْسِنٌ	
مَوَاتٍ	مُؤْمِنٌ	يَدٌ	صَاحِبٌ	لُبٌ	
يَتْلُو	عَمَلٌ	رَسُولٌ	آيَةٌ	تَوَابٌ	

10 نمبر

سوال 5 : آیات کو ان کے ترجمے سے لکیر کے ذریعے ملائیں۔

Match the verses with lines to the translation.

ہم صرف تیری ہی عبادت کرتے ہیں اور ہم
صرف تجھ سے ہی مدد مانگتے ہیں۔

1 وَلَا تَلْبِسُوا الْحَقَّ بِالْبَاطِلِ وَتَكْتُمُوا
الْحَقَّ وَأَنْتُمْ تَعْلَمُونَ۔

اس نے کہا میں اللہ کی پناہ میں آتا ہوں کہ میں
جاہلوں میں سے ہو جاؤں۔

2 إِيَّاكَ نَعْبُدُ وَإِيَّاكَ نَسْتَعِينُ۔

(یہ) وہ کتاب ہے جس میں کوئی شک نہیں،
پرہیزگاروں کے لیے ہدایت ہے۔

3 قَالَ أَعُوذُ بِاللَّهِ أَنْ أَكُونَ مِنَ الْجَاهِلِينَ

اور حق کو باطل کے ساتھ خلط ملط نہ کرو، اور حق
کو (ن) چھپاؤ حالانکہ تم جانتے ہو۔

4 قُلْ لِلَّهِ الْمَشْرِقُ وَالْمَغْرِبُ

آپ کہہ دیجئے (کہ) مشرق اور مغرب اللہ ہی کے
لیے ہے۔

5 ذَلِكَ الْكِتَابُ لَا رَيْبَ فِيهِ هُدًى
لِّلْمُتَّقِينَ۔